SET B – Module 3

EDGESENSE: FOREST EDGE PRESSURE & FRAGMENTATION ANALYSIS

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PROBLEM STATEMENT & HYPOTHESIS

Forest fragmentation caused by human activities such as roads, settlements and agriculture increases the proportion of forest edges relative to core forest areas. We hypothesize that these ***edge regions experience higher ecological stress due to greater exposure*** *to human disturbance, microclimatic changes, and habitat degradation.*

Therefore, ***quantifying forest edge pressure and fragmentation using satellite imagery*** *is essential to identify vulnerable forest patches and assess ecological stress patterns across the landscape.*

LITERATURE SURVEY

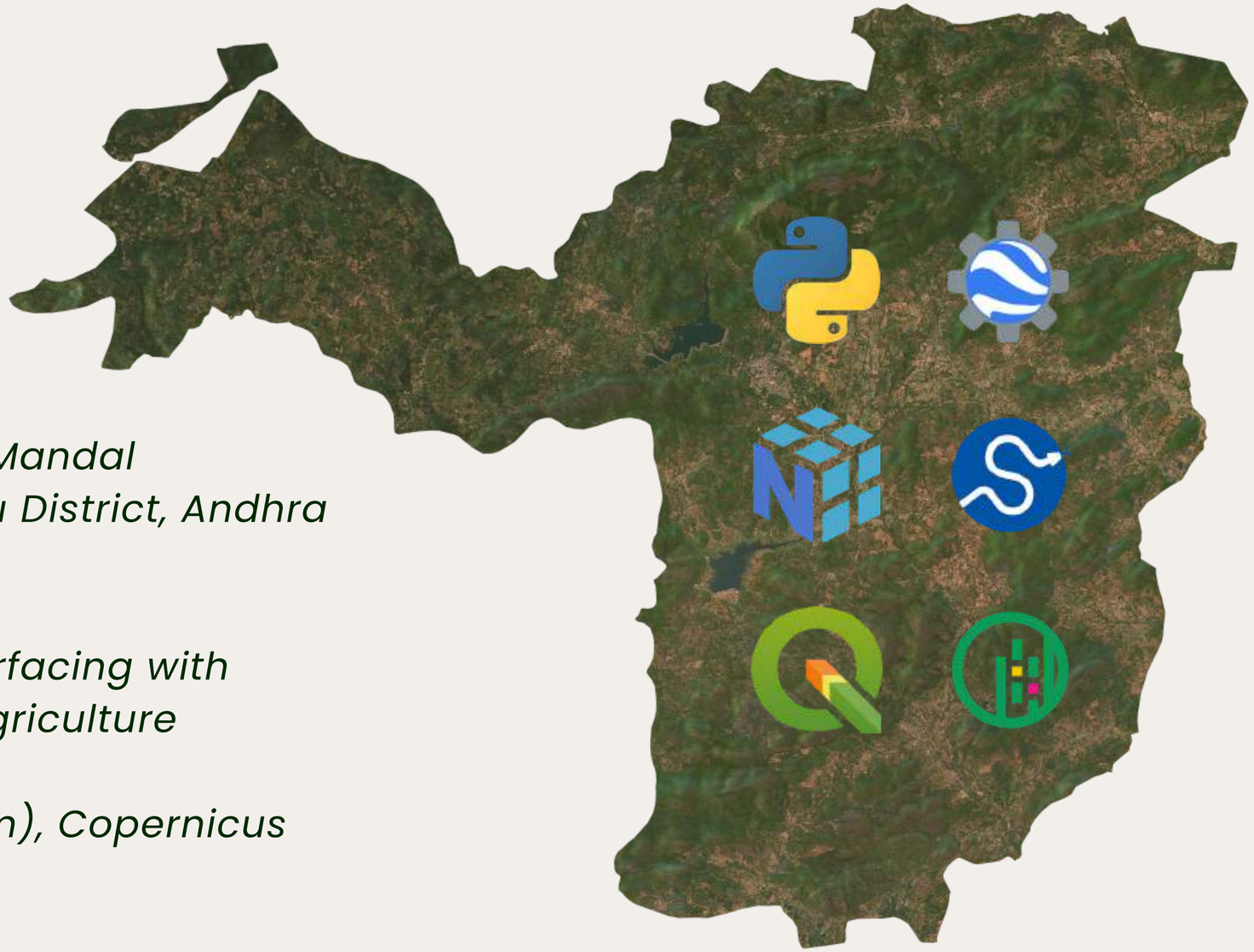
- ***Habitat fragmentation, variable edge effects, and the landscape-divergence hypothesis***.*Laurance et al. (2007) - Used field plots at different distances from forest edges in the Amazon to show that areas near edges are ecologically unstable. Gives us the scientific basis that edge-to-core ratio is a valid proxy for forest stress.*

- ***Forest fragmentation and edge effects from deforestation and selective logging in the Brazilian Amazon*** *Cochrane & Laurance (2008) - Used Landsat imagery + GIS buffers to measure fragmentation across the Brazilian Amazon. First to formally define and measure edge zones using fixed-distance dilation.*

- ***National assessment of forest fragmentation in India: Landscape indices as measures of the effects of fragmentation and forest cover change.*** *Reddy et al. (2013) - Used FRAGSTATS indices on satellite data across all of India's forest zones. Found Eastern Ghats has the lowest core forest % among non-arid zones.*

- ***Assessment of fragmentation and disturbance patterns in Eastern Ghats: A case study in R.V. Nagar Range, Visakhapatnam District.***Reddy et al. (2011) - Studied forest loss in the Eastern Ghats using multi-year IRS imagery and road-buffer threat mapping. Confirms the same stress drivers (roads, settlements) are active in our study area.

STUDY AREA & TECH STACK



- **Study Area:** *Rampachodavaram Mandal (Subdistrict), Alluri Sitharama Raju District, Andhra Pradesh*

- **Characteristics:** *Mixed forest interfacing with human settlements, roads, and agriculture*

- **Data Sources: Sentinel-2** *MSI (10m), Copernicus Global Land Cover*

- **Tech Stack:** *Python (Rasterio, GeoPandas, NumPy, SciPy), QGIS 3.28+ API, Google Earth Engine (GEE)*

Subdistrict: ***Rampa Chodawaram***
Year: ***2018***
Boundary Source: ***Survey of India***

PIPELINE OVERVIEW

Data Acquisition & Preprocessing (GEE)

PIPELINE OVERVIEW

Data Acquisition & Preprocessing (GEE) ➡ **Ecological Index Computation (NDVI & NDMI)**

PIPELINE OVERVIEW

Data Acquisition & Preprocessing (GEE) ➡ **Ecological Index Computation (NDVI & NDMI)** ➡ **Binary Forest & Road Mask Generation**

PIPELINE OVERVIEW

Data Acquisition & Preprocessing (GEE) → **Ecological Index Computation (NDVI & NDMI)** → **Binary Forest & Road Mask Generation** → **Spatial Edge/Core Separation**

PIPELINE OVERVIEW

Data Acquisition & Preprocessing (GEE) → Ecological Index Computation (NDVI & NDMI) → Binary Forest & Road Mask Generation → Spatial Edge/Core Separation → Fragmentation Metrics & Change Analysis

STEP 1: AUTOMATED SATELLITE DATA RETRIEVAL

METADATA:

- **Administrative Boundary:** Survey of India - Rampa Chodavaram Subdistrict
- **Satellite Imagery:** Sentinel-2 Multispectral Imagery (ESA Copernicus Program)
- **Bands:** *Red (Band 4), NIR (Band 8), Narrow NIR (Band 8A), SWIR (Band 11), Green (Band 3)*
- **Duration:** 2018-2024

RETRIEVAL STEPS:

- Fetch multi-year Sentinel-2 imagery using **Google Earth Engine API**
- **Clip** to AOI
- Search: Post-monsoon with relaxed cloud cover thresholds (1% to 50% with step size=5)
- Create **median composite**
- **Downloads data in tiled grid** - to avoid download size limits on GEE
- Mosaic the tiles to get a single .tif file
- omnicloudmask used to create cloud mask and **remove cloudy areas**, if any.

DISTRICT BASEMAP

2018

2019

2020

2021

2022

2023

2024

CLOUD MASKS

2018

2019

2020

2021

2022

2023

2024

STEP 2: SPECTRAL INDEX COMPUTATION

- Red, NIR, Narrow NIR and SWIR bands extracted from Sentinel-2 GeoTIFF and normalised (divided by 10,000 to convert to reflectance)

- NDVI & NDMI computed using the standard formulas:
  NDVI = (NIR - Red) / (NIR + Red)
  NDMI = (Narrow NIR - SWIR) / (Narrow NIR + SWIR)

- Values clipped to [-1, 1] & nodata pixels (out of subdistrict boundary) masked out

- Output: per-year(2018-2024) NDVI & NDMI raster saved as GeoTIFF

WHY USE BOTH NDVI & NDMI?

NDVI:

- Measures vegetation **greenness** / photosynthetic activity
- Helps detect **presence of vegetation**
- Limitation: **Cannot clearly distinguish forest from crops or grassland**

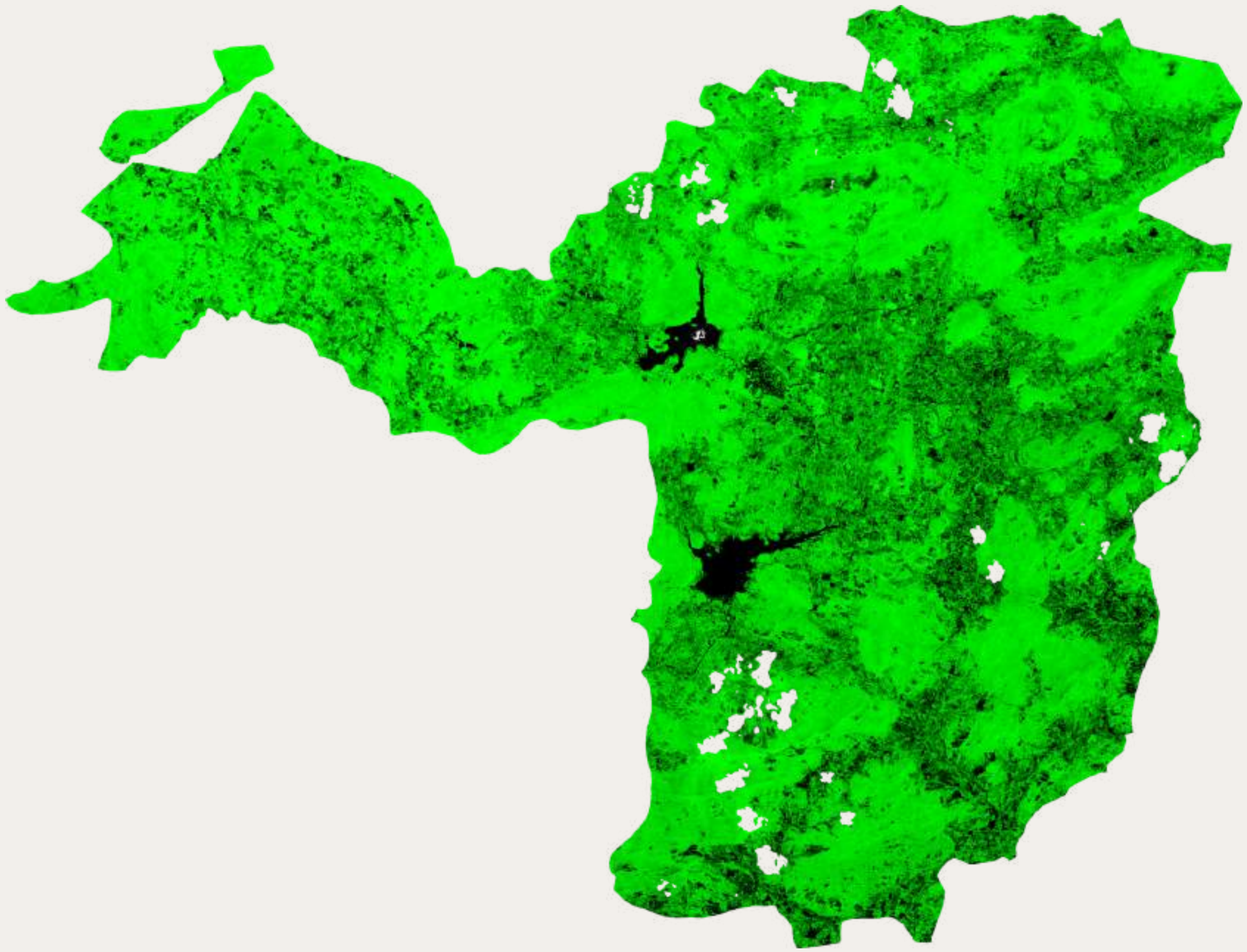
NDMI:

- Measures vegetation **moisture** content
- Helps identify **dense, moist vegetation** like forests
- Useful for detecting vegetation health and water stress

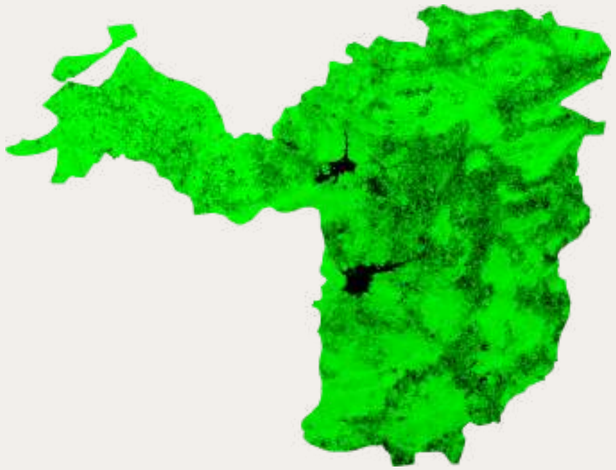
Both Together

- NDVI → detects vegetation presence
- NDMI → detects vegetation moisture
- Forests usually have high greenness + high moisture
- Combining both ***reduces false detection of crops or dry vegetation***
- Produces a ***more accurate*** binary forest mask

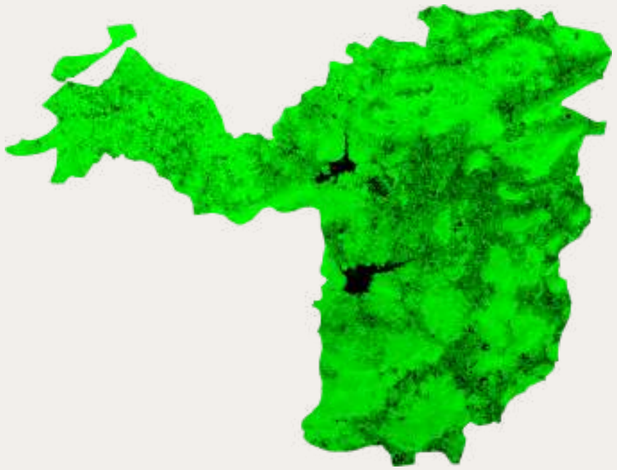
NDVI OUTPUTS



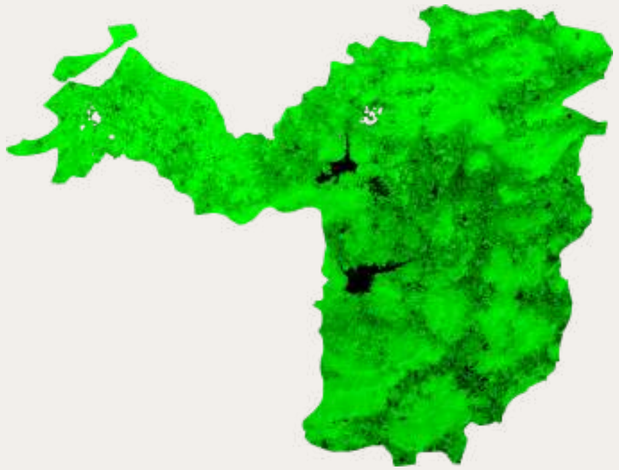
2018



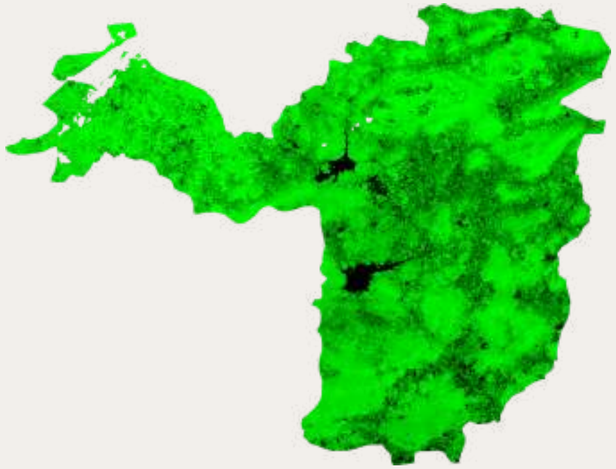
2019



2020



2021



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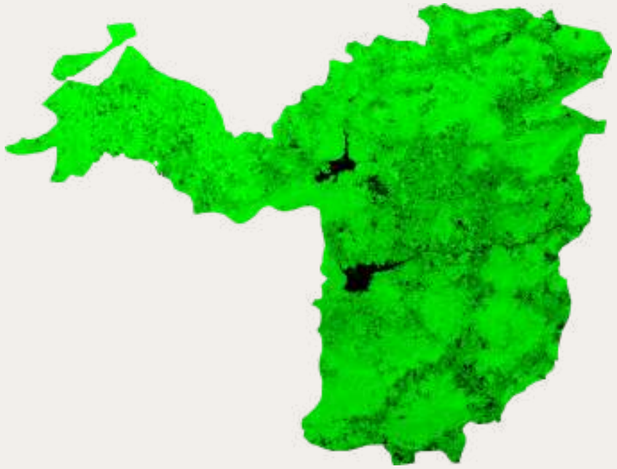

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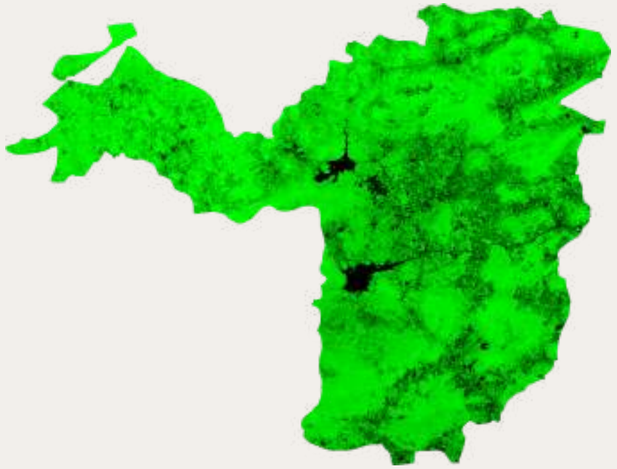
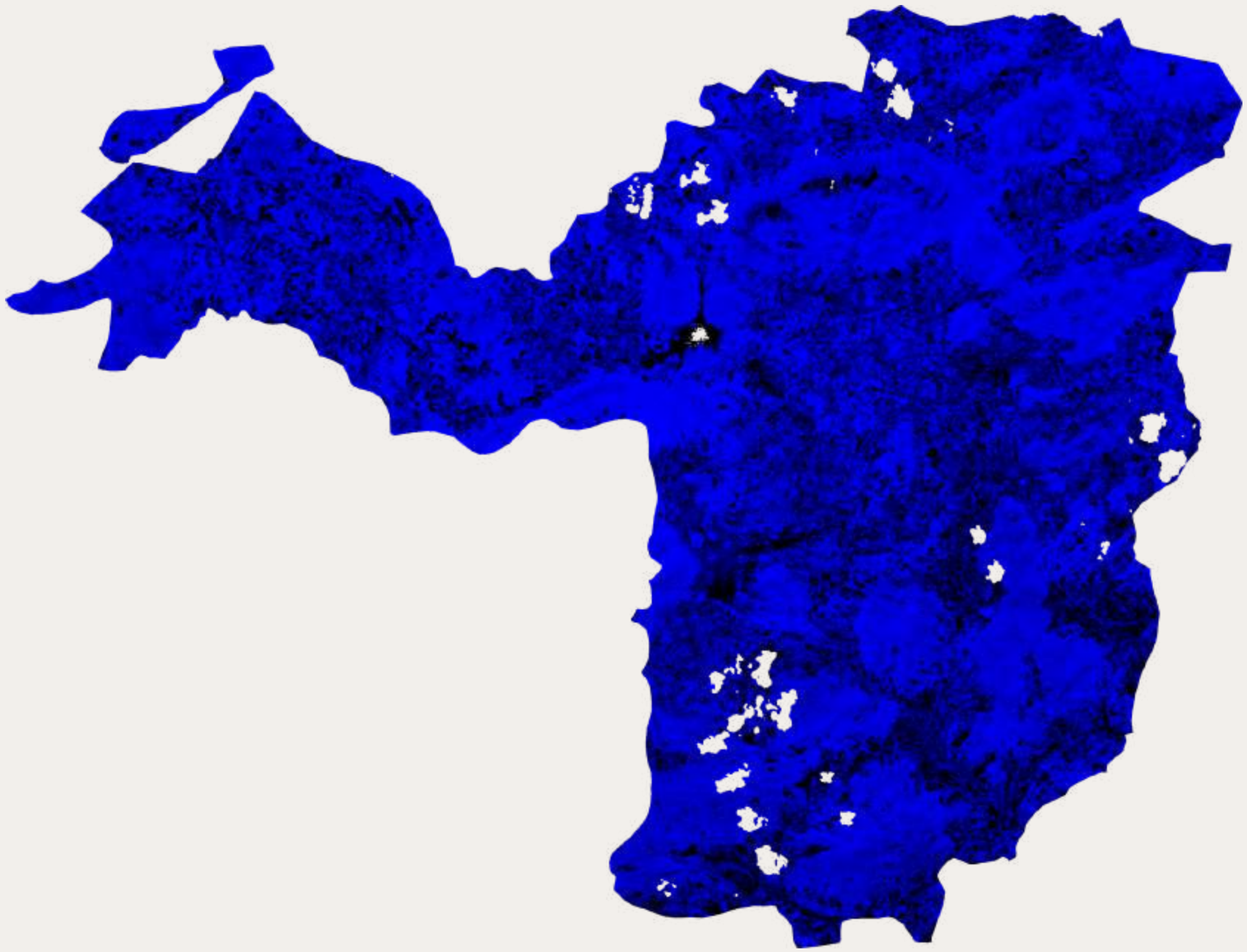


2024

NDMI OUTPUTS



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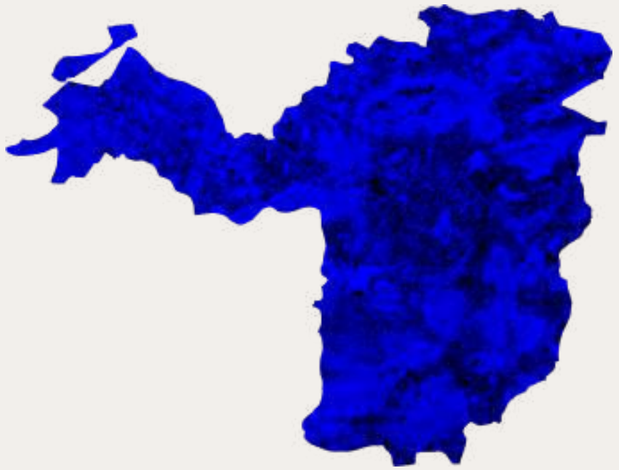

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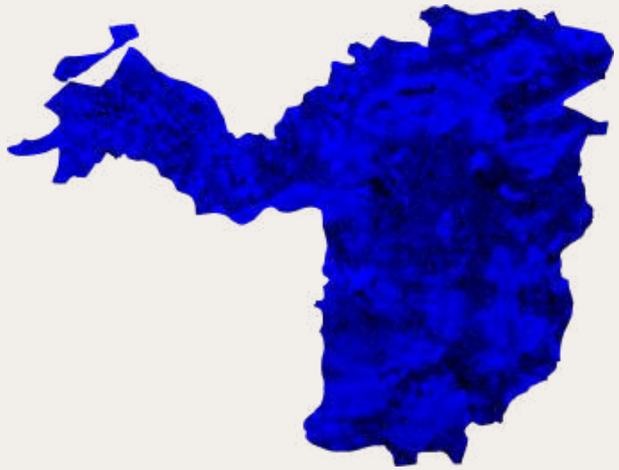
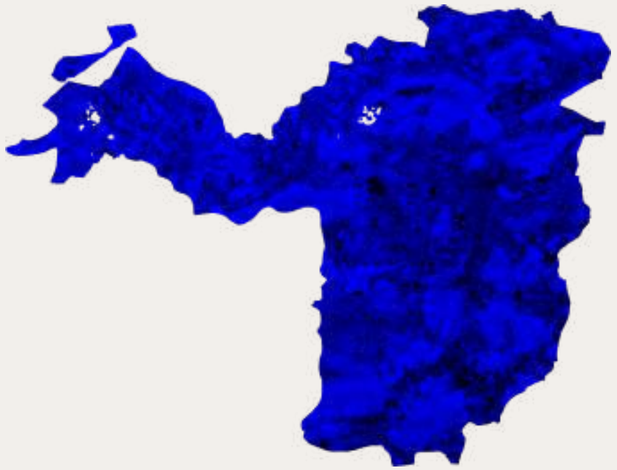


2020



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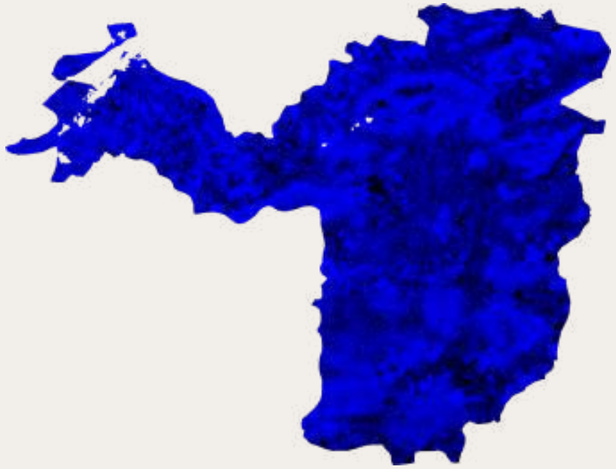

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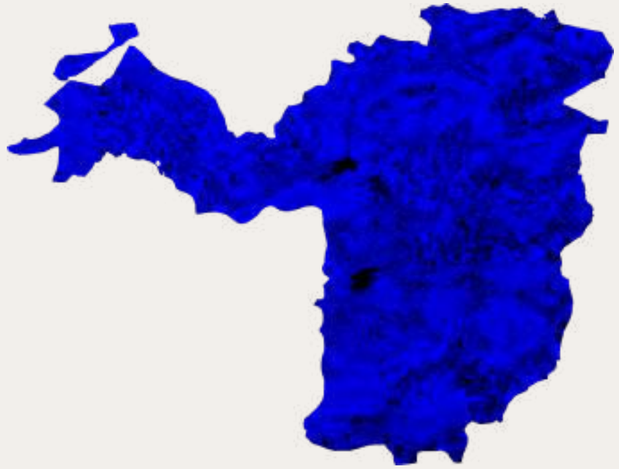

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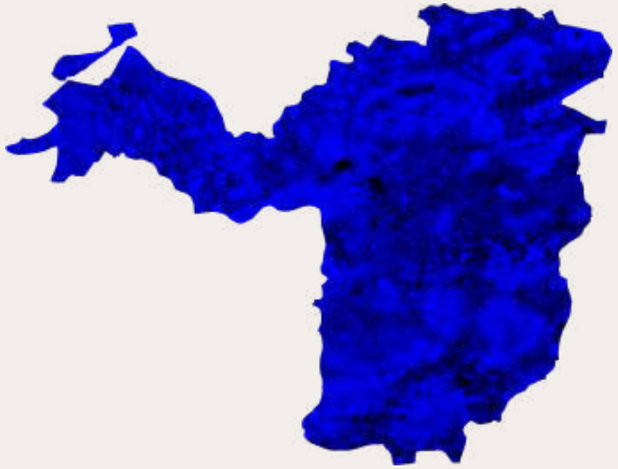
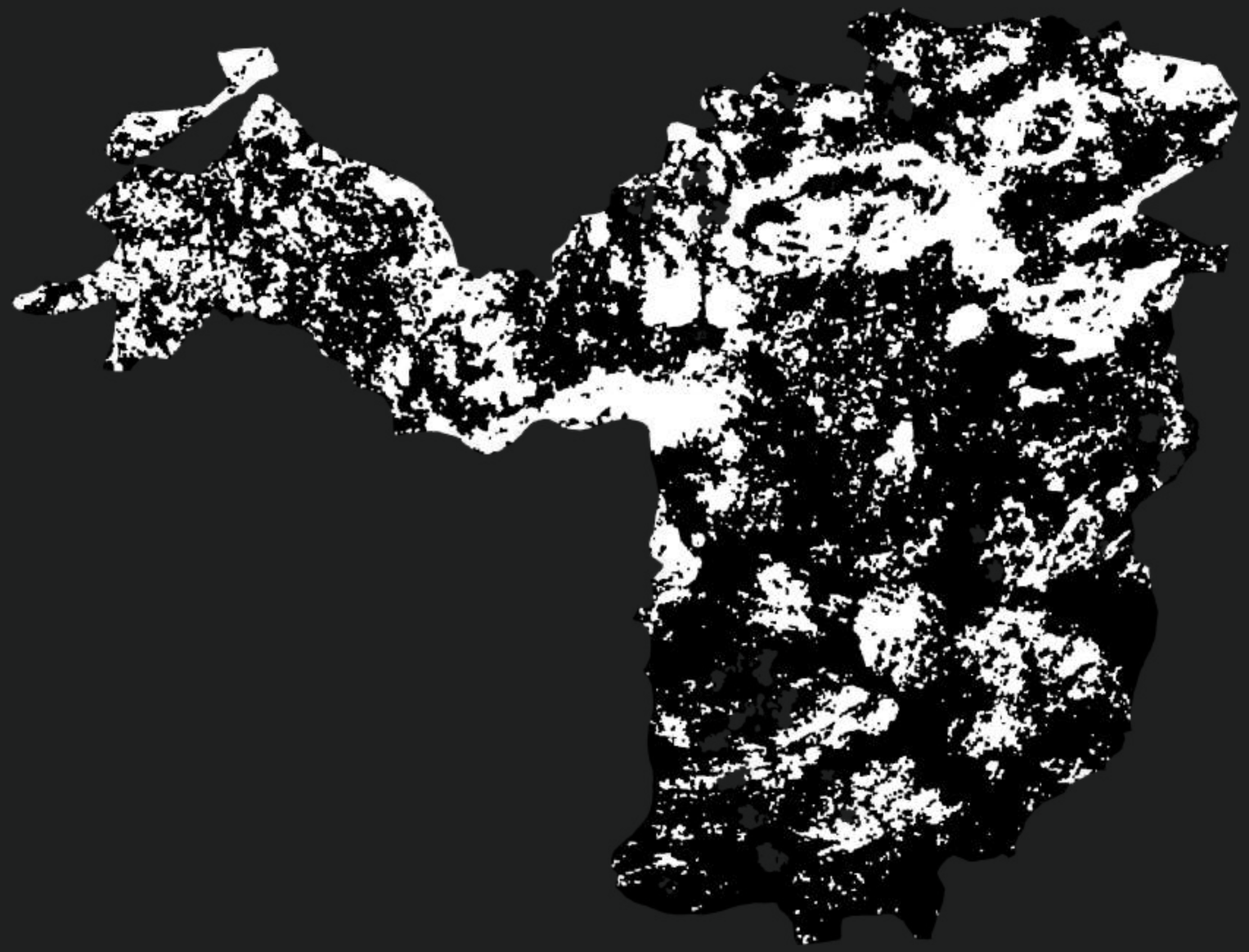


2024

STEP 3A: FOREST MASK GENERATION

- *A threshold-based classification*
- *Pixels with **NDVI ≥ 0.65 and NDMI ≥ 0.3** classified as Forest (1), all others as Non-Forest (0)*
- *Output: binary forest mask GeoTIFF per year*
- ***Morphological Smoothing:** Utilised scipy.ndimage to apply structural opening and closing operations. This removes isolated "noisy" pixels and fills small canopy gaps, creating more realistic, cohesive forest patches*

FOREST MASKS



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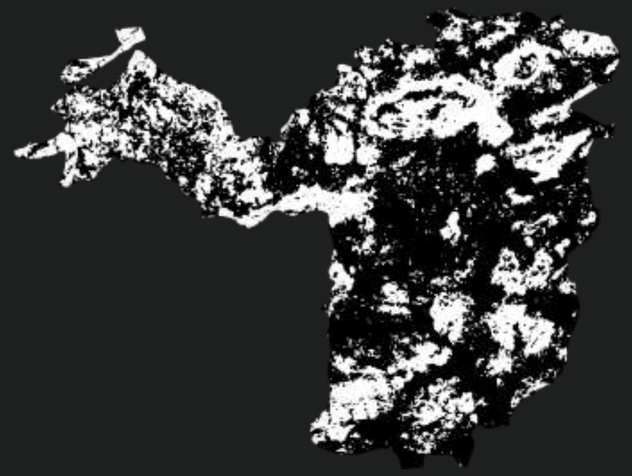

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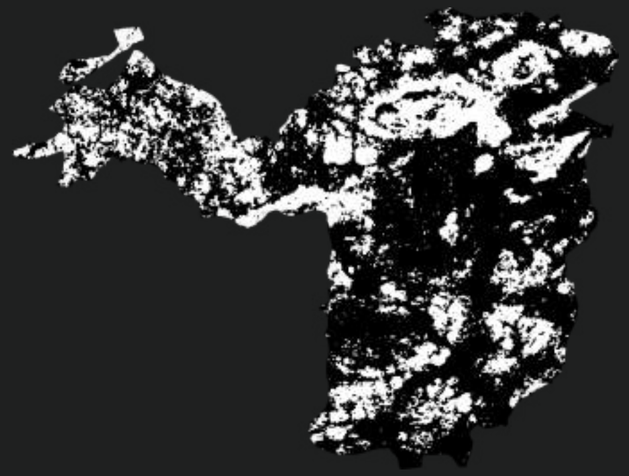

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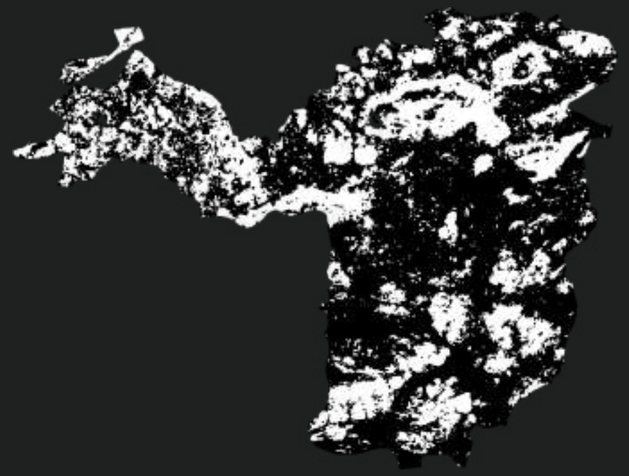
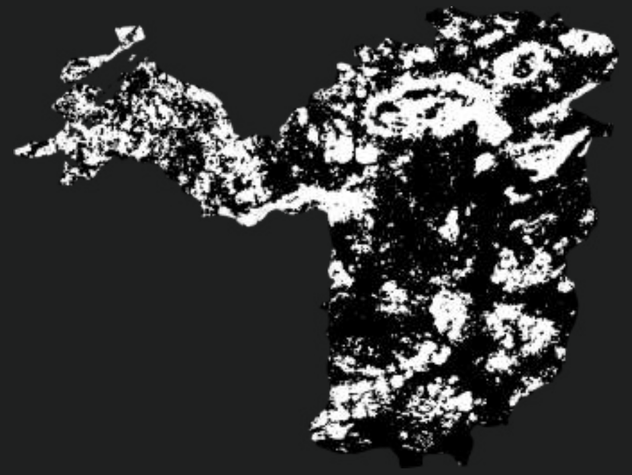


2021



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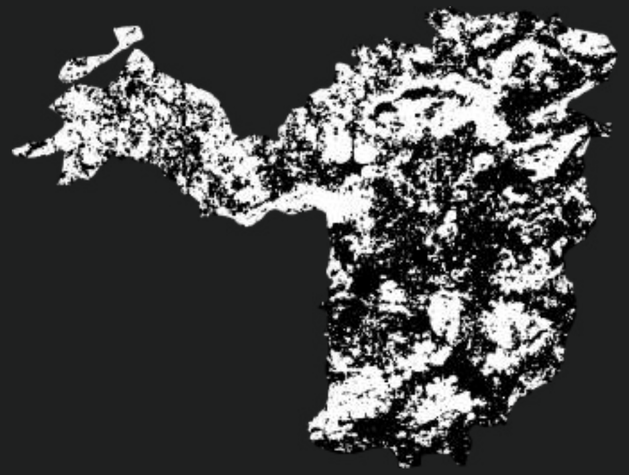
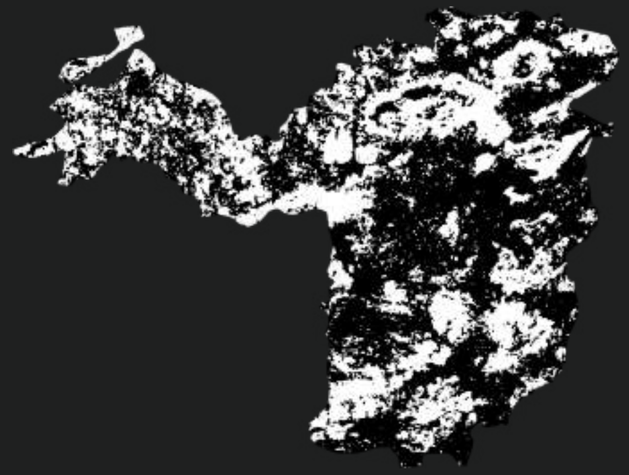
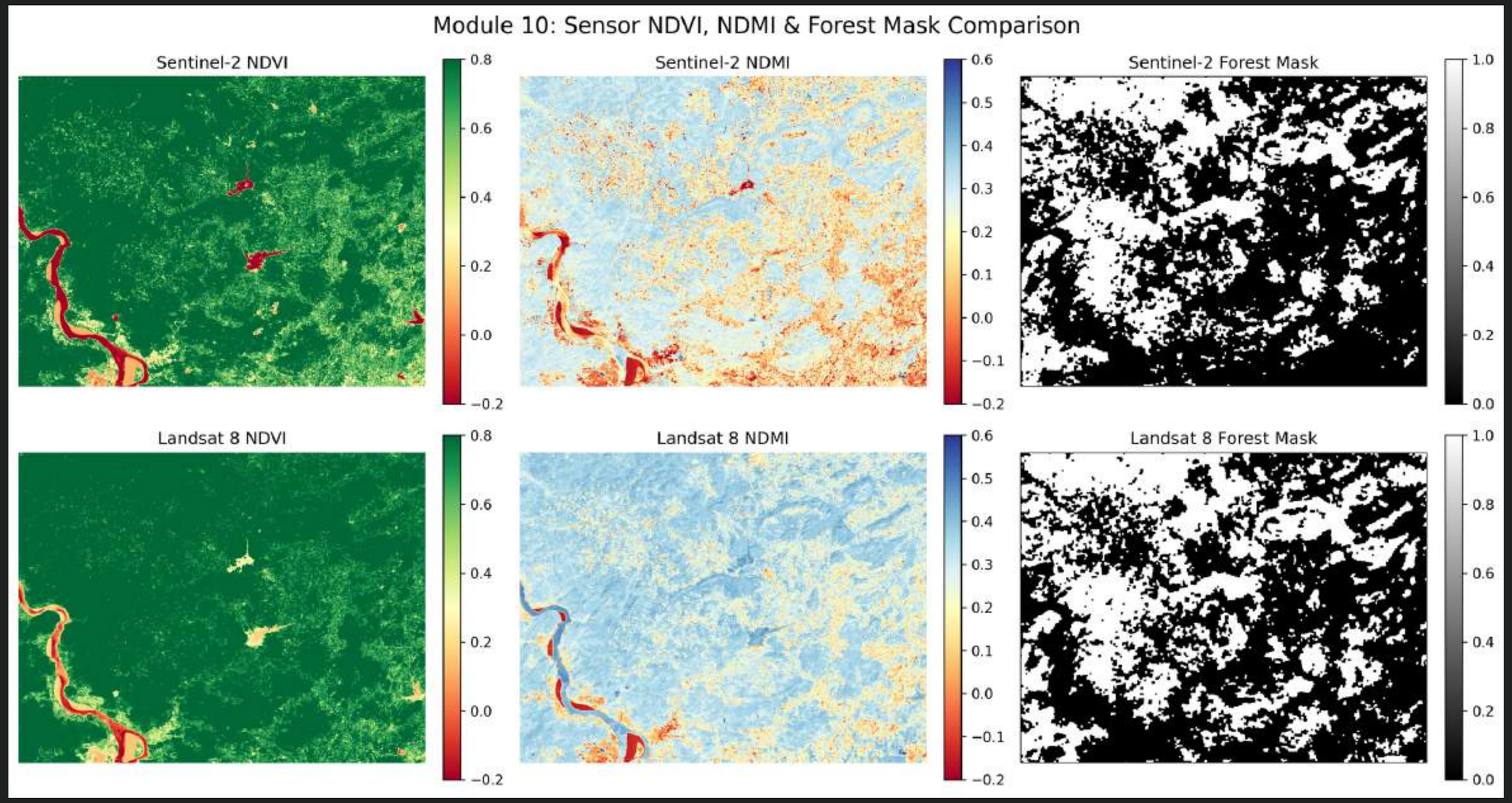


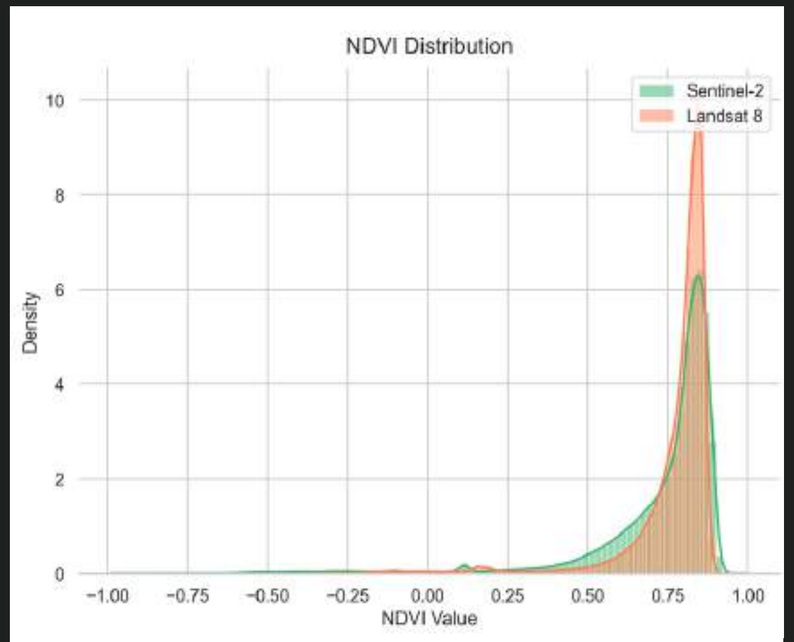
2023

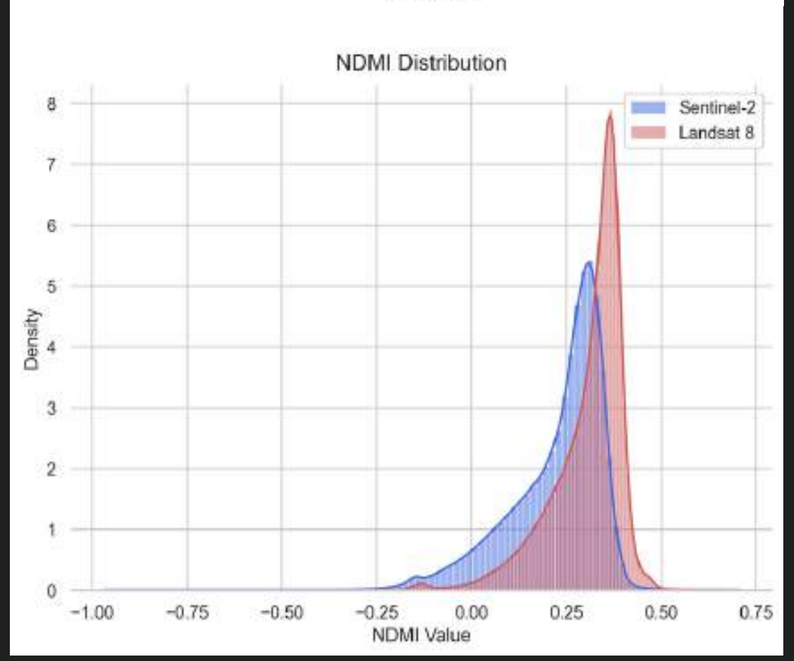


2024

INDEX VALIDATION







INDEX VALIDATION

Metric	Value
NDVI Correlation (r)	0.9033
NDMI Correlation (r)	0.7396
Forest Mask Agreement	88.25%

- *S2 - lower NDVI/NDMI - narrower bands and higher resolution - more precise, less mixed signals, making values more conservative. L8 - broader bands and coarser resolution - inflate values and saturates earlier - cannot distinguish well between dense vegetation levels.*

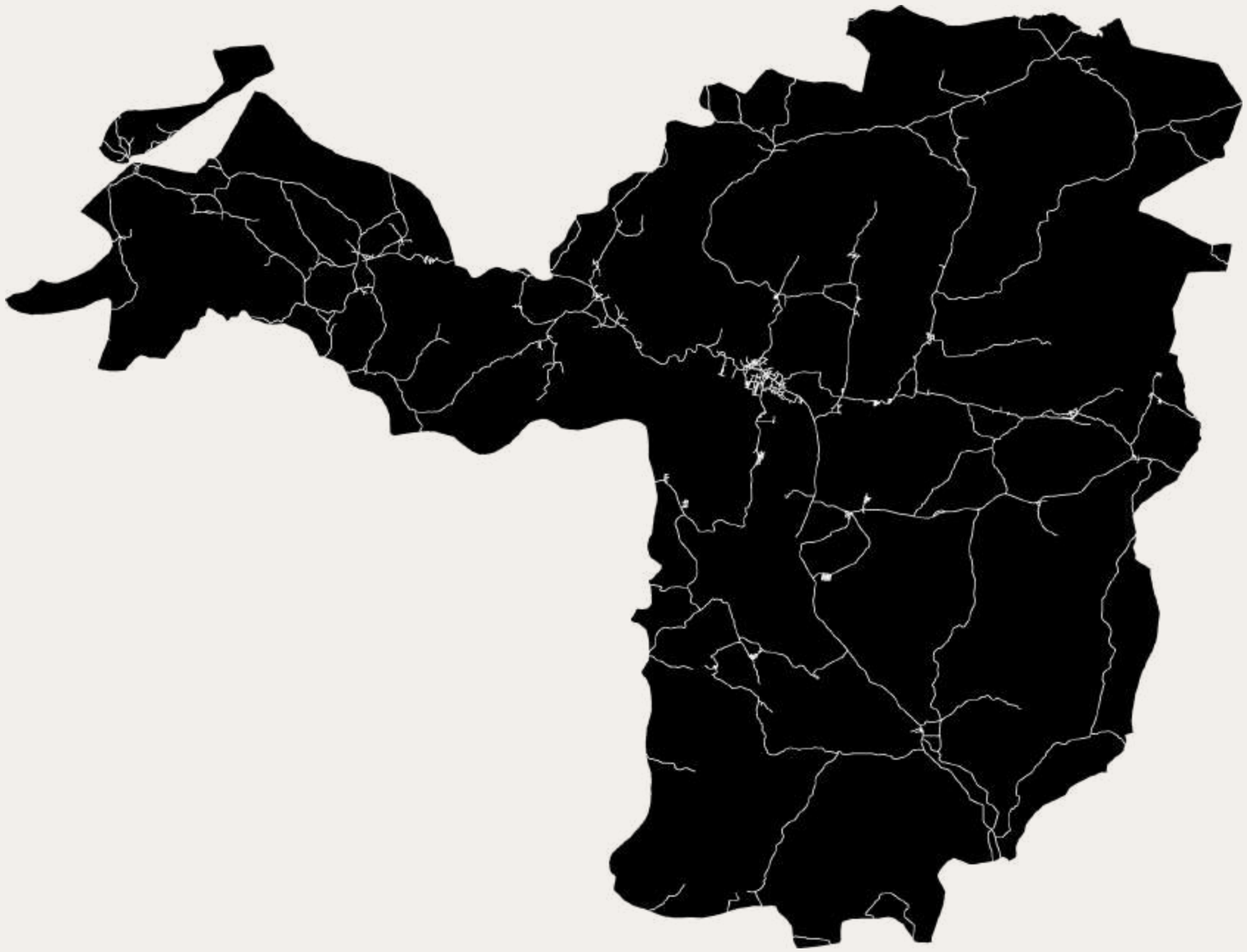
BANDS USED

Satellite	Band Type	Band Number	Wavelength (micro m)
Landsat-8	Red	4	0.636 – 0.673
	NIR	5	0.851 – 0.879
	SWIR 1	6	1.566 – 1.651
Sentinel-2	Red	4	~0.665
	NIR	8	~0.842
	SWIR 1	11	~1.610

STEP 3B: ROAD MASK GENERATION

- *Roads fetched from OpenStreetMap using Overpass API within AOI*
- *Road centerlines buffered (in meters) to approximate real road width*
- *Buffered roads converted to binary mask: Road (1), Non-road (0)*
- *Mask aligned to reference grid and saved as GeoTIFF per year*

ROAD MASK

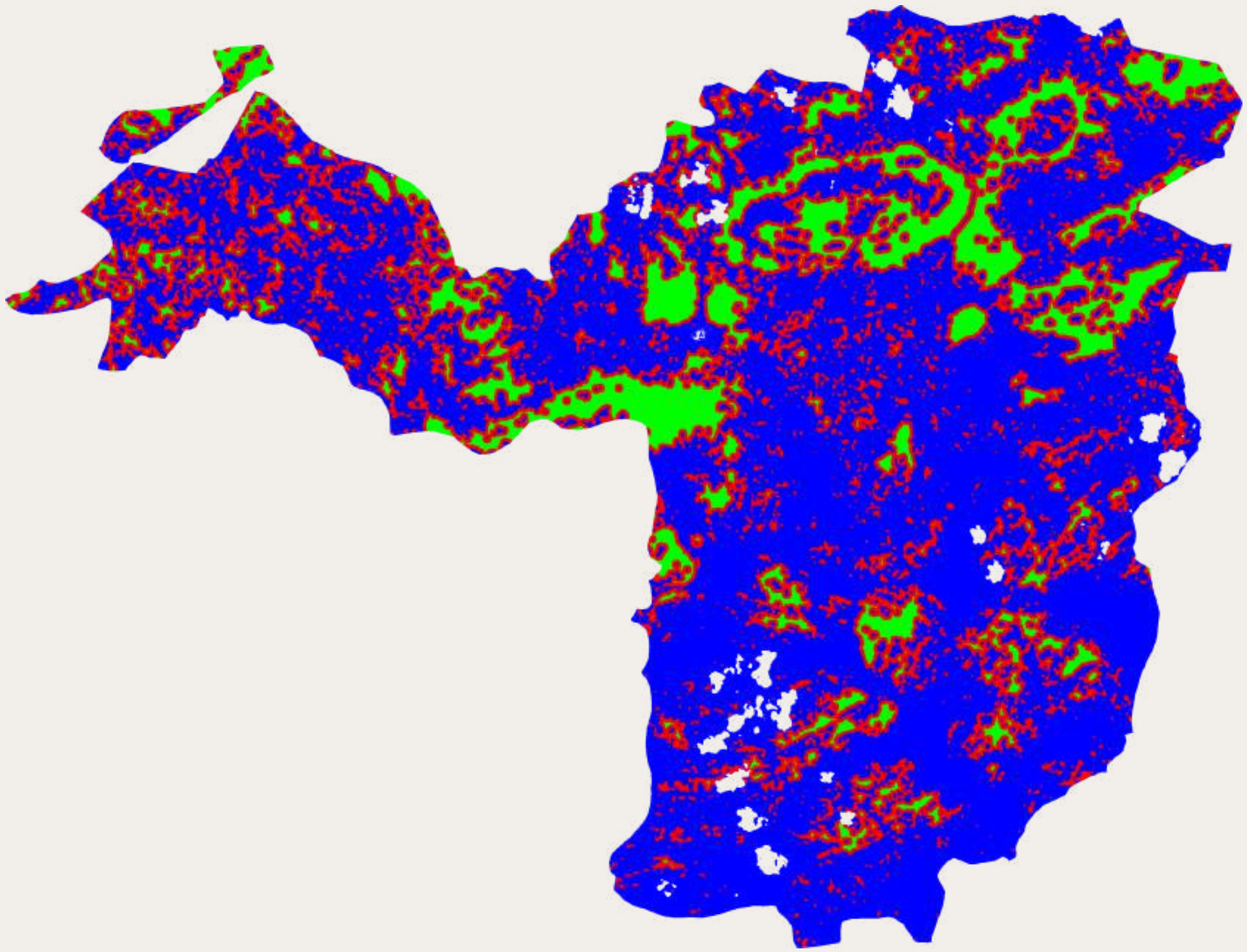


*Did not change much over the years

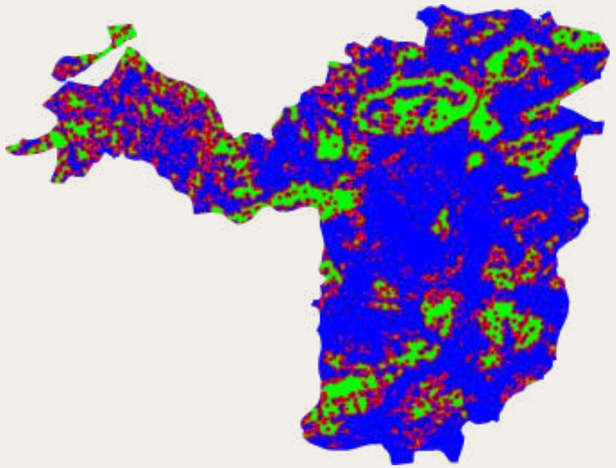
STEP 4: EDGE & CORE SEPARATION

- ***Road-aware Forest Adjustment:*** *Forest pixels overlapping roads are reclassified as Non-Forest using the road mask*
- ***Neighbourhood-based Edge Detection:*** *Circular structural kernel defines the edge depth (in pixels)*
- ***Spatial Expansion of Non-Forest:*** *Binary dilation applied to non-forest areas (including roads) to push edges into forest*
- ***Edge & Core Classification:*** *Overlap → Edge, remaining interior forest → Core (RGB: Edge–Red, Core–Green, Non-Forest–Blue)*

EDGE-CORE MASKS



2018



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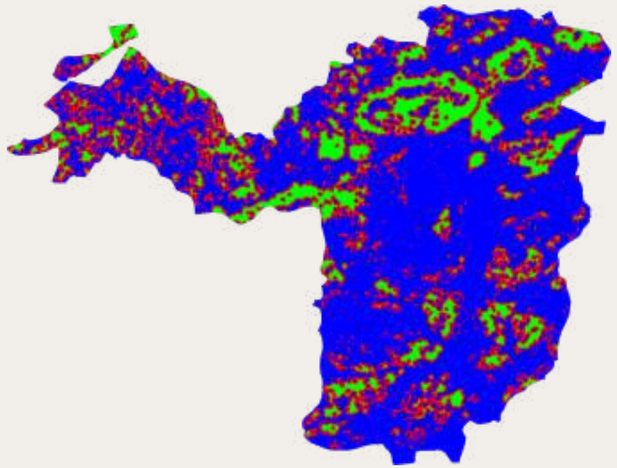
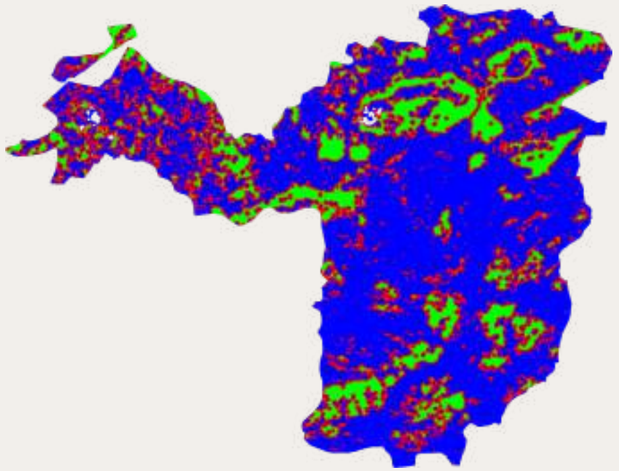


2020



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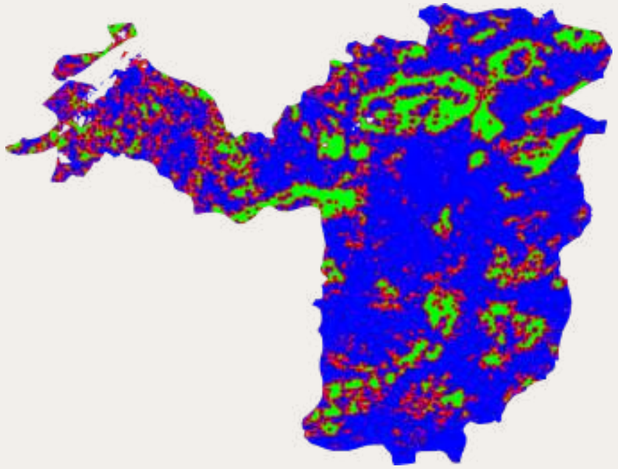
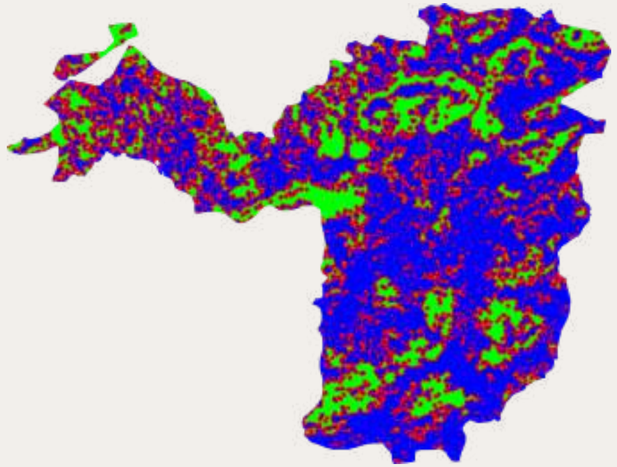


2022



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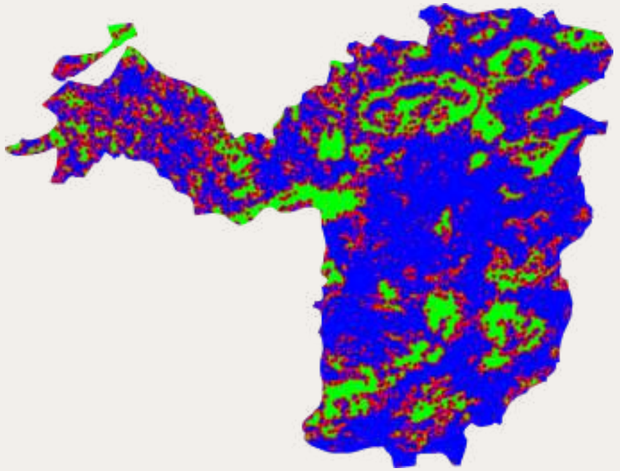


2024

STEP 5: QUANTIFYING FOREST HEALTH: KEY METRICS

1. ***Total Forest Area:*** *The overall footprint of all forest pixels combined.*

2. ***Number of Forest Patches:*** *The total count of distinct, isolated forest fragments (filtered to >= 0.5 ha).*

3. ***Mean Patch Size:*** *The average area of individual forest patches, indicating whether the remaining forest is breaking into smaller pieces.*

4. ***Mean Core Area Fraction:*** *The proportion of sheltered interior habitat relative to total patch area.*

STEP 5: QUANTIFYING FOREST HEALTH: KEY METRICS

5. ***Mean Patch Cohesion:*** *A measure of spatial compactness from 0 to 1, where lower values indicate more elongated or disaggregated geometry.*

6. ***Edge:Core Ratio:*** *Our primary hypothesis variable; a higher ratio indicates a larger proportion of vulnerable edge area compared to protected core.*

7. ***Mean Shape Index:*** *A measure of perimeter complexity (>= 1), where a higher value signifies a more irregular, jagged boundary.*

8. ***Stress Pressure Index:*** *Our custom composite ecological stress score, calculated by multiplying the Edge:Core Ratio by the Shape Index.*

COMPARING CUSTOM METRICS TO FRAGSTAT METRICS

- **Practical limitations of 'pylandstats':** It **didn't integrate cleanly** into our existing geospatial **pipeline**, requiring tedious I/O steps and format conversions. Moreover, **intermediate steps are opaque.** It is difficult to inspect or visualise patch labeling, edge dilation, or per-pixel assignments at each stage of the analysis.

- **Methodological differences:** Our custom **shape index formula** uses a different reference standard (**circle vs. square**). Moreover, our **core area extraction** uses road-aware masking and circular-kernel dilation of non-forest into the forest, instead of a uniform buffer. So, **values** are **not directly interchangeable.**

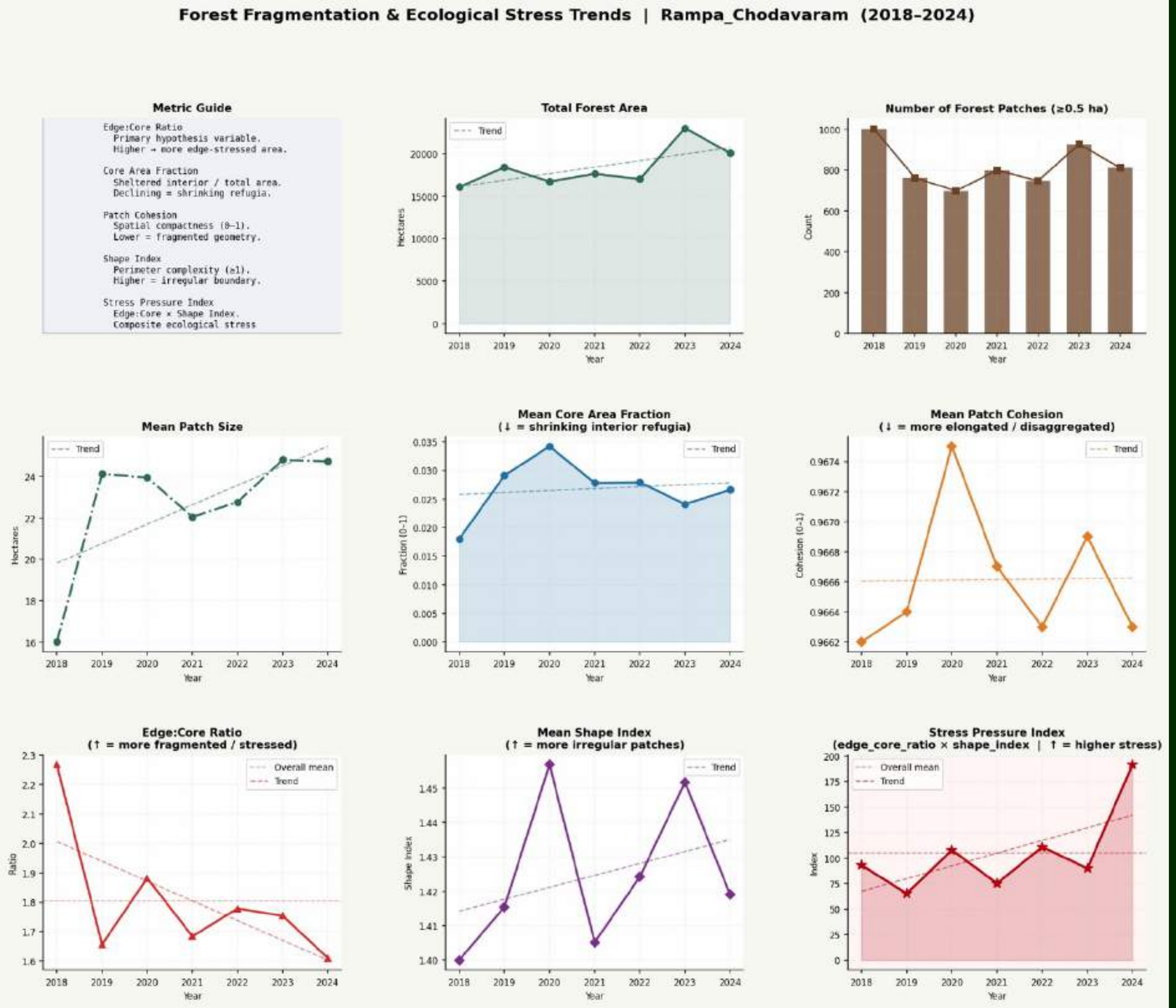
- **Consistent Landscape-Level Outputs:** Despite this, patch counts, total forest area, mean patch size, and shape index rankings aligned well across both methods. The two pipelines agree closely enough to give confidence that our **custom** implementation is **capturing** fragmentation **patterns faithfully**.

year	custom_n_patches	custom_total_forest_ha	custom_mean_patch_ha	custom_largest_patch_ha	custom_mean_shape_index	custom_mean_core_area_fraction	custom_total_edge_core_ratio
2018	1003	16067.86	16.02	1698.11	1.3999	0.018	2.268
2019	762	18382.51	24.124	1827.78	1.4154	0.0291	1.6564
2020	698	16715.86	23.948	1870.64	1.4569	0.0342	1.8822
2021	799	17617.2	22.049	1790.95	1.4051	0.0278	1.684
2022	747	17002.32	22.761	1833.77	1.4242	0.0279	1.7785
2023	927	22991.58	24.802	1939.28	1.4518	0.0241	1.7536
2024	811	20051.67	24.725	1846.51	1.419	0.0266	1.6101

Custom Metrics

year	fs_n_patches	fs_total_forest_ha	fs_mean_patch_ha	fs_largest_patch_ha	fs_mean_shape_index	fs_mean_core_area_fraction	fs_total_edge_core_ratio
2018	993	15232.01	15.339	1610.704	1.4207	0.021	1.9723
2019	750	17430.65	23.241	1732.117	1.4401	0.033	1.4456
2020	673	15840.61	23.537	1772.6	1.4835	0.04	1.6263
2021	782	16699.21	21.354	1696.666	1.4284	0.0322	1.4708
2022	729	16115.32	22.106	1737.337	1.4488	0.0326	1.5487
2023	917	21800.99	23.774	1837.512	1.4813	0.0276	1.5314
2024	794	19006.71	23.938	1749.335	1.445	0.0303	1.4135

Fragstat Metrics



TRENDS OBSERVED

WHY DO TRENDS BEHAVE THIS WAY?

Disturbance
(2018–2019)

- **Podu (shifting cultivation)** by tribal communities in Rampachodavaram is a seasonal cycle of clearing and burning small forest plots creating a high number of small, fragmented patches with irregular, jagged boundaries (Reddy et al., 2011).
- **Road expansion & settlement pressure** at forest margins is the dominant driver of fragmentation in Eastern Ghats AP — documented from 1988 onwards in this exact landscape (Reddy et al., 2011).
- **Result:** ~1000 patches, small mean size (~16 ha), high Edge:Core ratio (~2.3), irregular patch shapes, low cohesion , all consistent with metrics

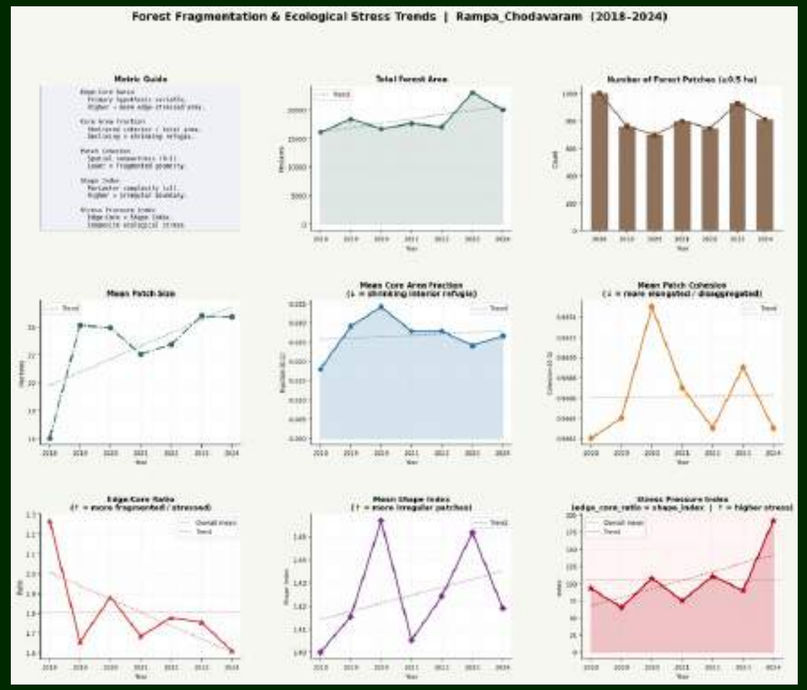
COVID19 Pause & Recovery
(2020–2021)

- **COVID-19 national lockdown (Mar 2020)** sharply reduced movement into forest areas across India. Podu activity, fuelwood collection, and encroachment all temporarily reduced , allowing vegetation on recently cleared plots to begin regrowing.
- **ISFR 2021:** Andhra Pradesh recorded one of India's highest forest cover increases (2.22%), confirming a period of measurable forest gain coinciding with reduced disturbance pressure.
- **Result:** Forest area rises, patch count falls (patches merge), mean patch size and core area fraction both improve , the landscape structurally consolidates from 2019 to 2021.

Renewed Stress
(2022–2024)

- **Post-lockdown** activity resumes: human pressure returns to forests - podu , infrastructure projects, and settlement growth restart, pushing patch count back up (to ~900 in 2023) and fragmenting consolidated patches.
- **ISFR 2023 forest fires:** Andhra Pradesh reported 5,287 km² of forest affected by fires (Nov 2023–Jun 2024) the highest among affected states. Fire breaks up contiguous patches and increase edge exposure.
- **Result:** Edge:Core ratio rises again in 2023 (~2.3), Shape Index peaks (most irregular boundaries of the period), and Stress Pressure Index spikes to its highest value

REFERENCES- REDDY ET AL.(2011)-HTTPS://WWW.SCIENCEDIRECT.COM/SCIENCE/ARTICLE/ABS/PII/S0925857413004102 , ISFR(2021)FOREST REPORT , ISFR(2023) FOREST REPORT



WHY DO TRENDS BEHAVE THIS WAY?

Total forest area(inc) - Forest area grows due to natural regrowth - podu fallow plots (typically 2–3 year cycle) and AP government afforestation - ISFR 2021 confirms gained 2.22% forest cover. The 2024 dip reflects fire-induced canopy loss (ISFR 2023).

Number of patches - Drops in 2020 reflect patch merging during (COVID lockdown). The 2023 rise is driven by renewed podu activity and fire events fragmenting larger patches back into smaller units .

Mean patch size - General increase confirms that net patch merging outpaces fragmentation across the period. As the number of patches decrease we see that mean patch size increases

Mean core area frac - Core area peaks in 2020 when patches are much large and least disturbed. Subsequent decline mirrors patch re-fragmentation post-lockdown: as smaller patches re-form.

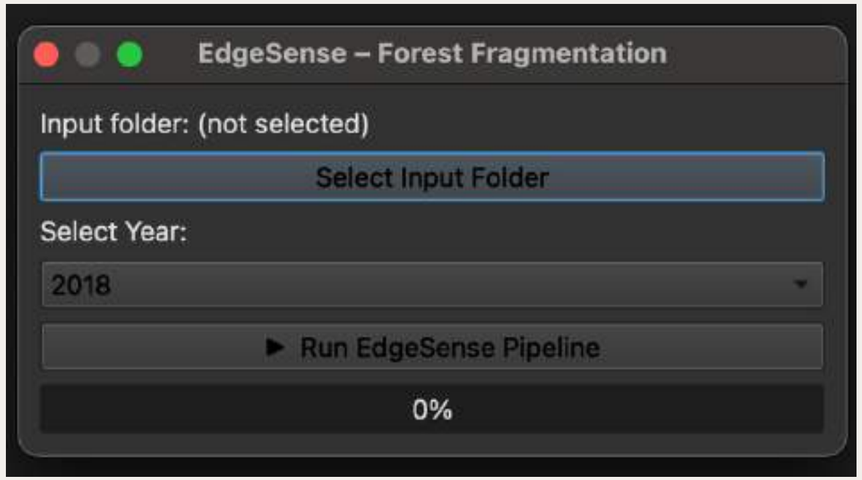
M**ean Patch cohesion** - Cohesion tracks spatial compactness peaks when patches are large and circular, falls when patches are re-fragmented by fire or podu into irregular shapes.

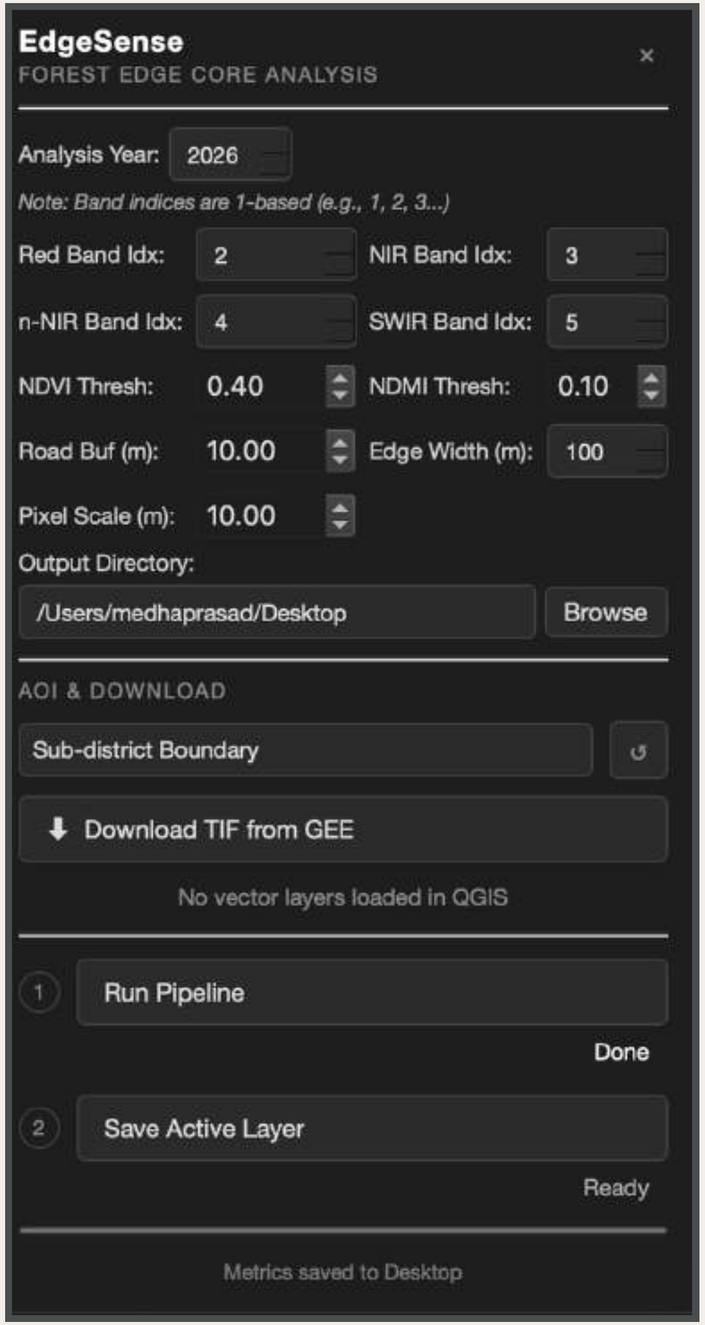
Edge core ratio - Overall declining trend confirms structural improvement - forests are gaining more interior area relative to edge.

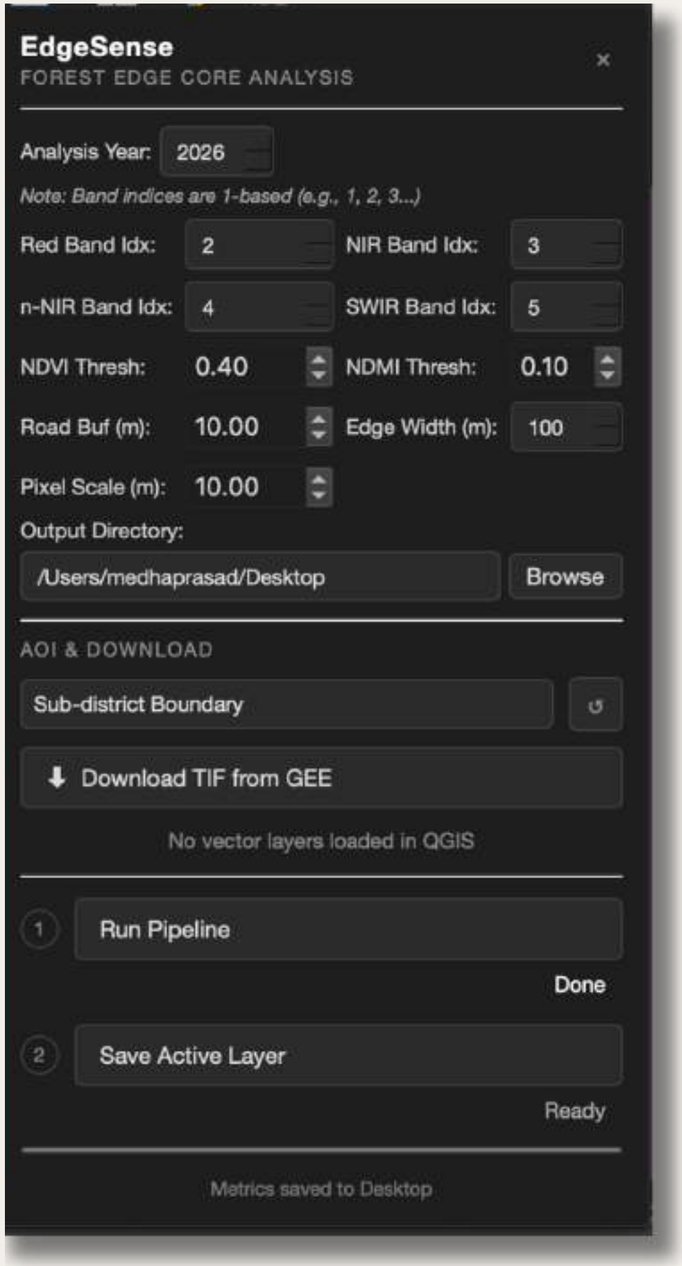
Mean shape index - Shape index measures boundary irregularity - rising means patches are becoming more jagged. fire scars generating non-circular patch boundaries.

Stress pressure index - Edge:Core × Shape Index. The rising trend capture stress increase despite area increase.

QGIS PLUGIN


EdgeSense – Forest Fragmentation
Input folder: (not selected)
Select Input Folder
Select Year:
2018
Run EdgeSense Pipeline
0%


EdgeSense
FOREST EDGE CORE ANALYSIS
Analysis Year: 2026
Note: Band indices are 1-based (e.g., 1, 2, 3...)
Red Band Idx: 2
NIR Band Idx: 3
n-NIR Band Idx: 4
SWIR Band Idx: 5
NDVI Thresh: 0.40
NDMI Thresh: 0.10
Road Buf (m): 10.00
Edge Width (m): 100
Pixel Scale (m): 10.00
Output Directory:
/Users/medhaprasad/Desktop
Browse
AOI & DOWNLOAD
Sub-district Boundary
Download TIF from GEE
No vector layers loaded in QGIS
1
Run Pipeline
Done
2
Save Active Layer
Ready
Metrics saved to Desktop



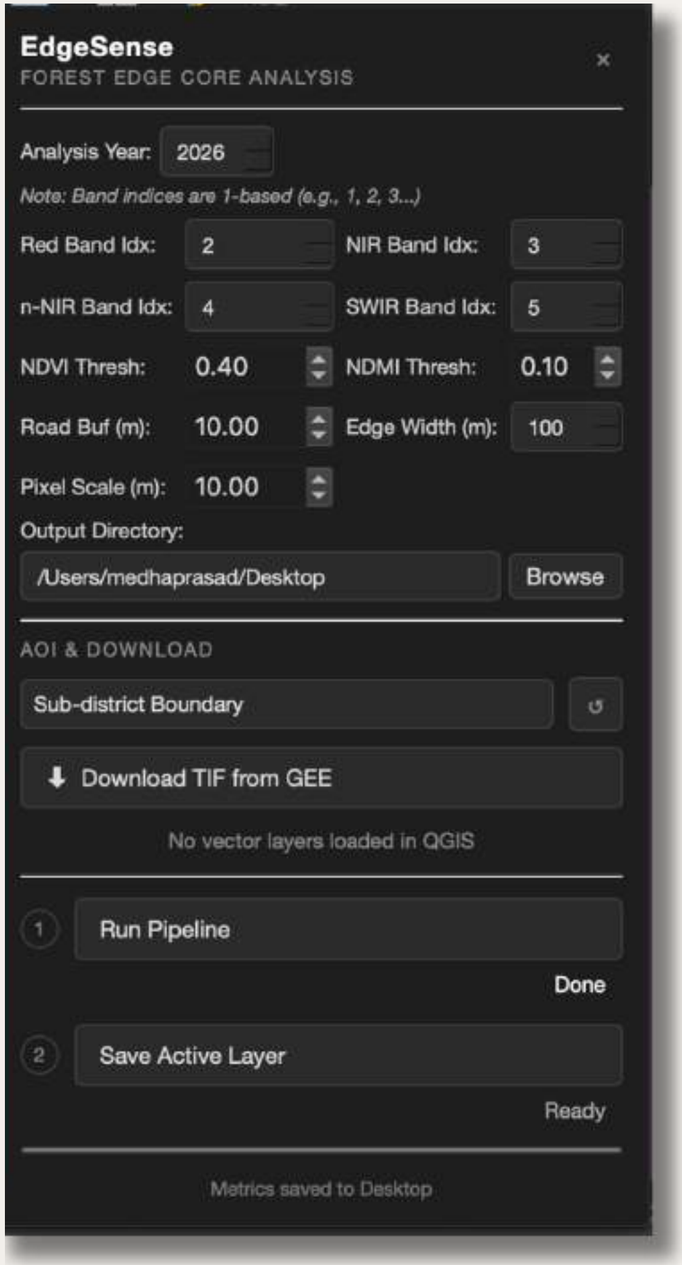
EDGESENSE

QGIS PLUGIN FOR FOREST FRAGMENTATION ANALYSIS

- **A Unified, Script-Free Environment:** The EdgeSense module provides a complete GUI for our analysis workflow. **No separate scripts need to be run outside the plugin.**

- **Ensuring Reproducibility:** By centralizing the workflow inside a custom QGIS interface panel, the **plugin ensures standardized processing, reproducible results, and immediate visualization.**

- **Modular Design:** As **part of the EcoLand-OS tool,** EdgeSense focuses on edge/core spatial metrics and **can easily integrate with other ecological stress modules.**



EDGESENSE

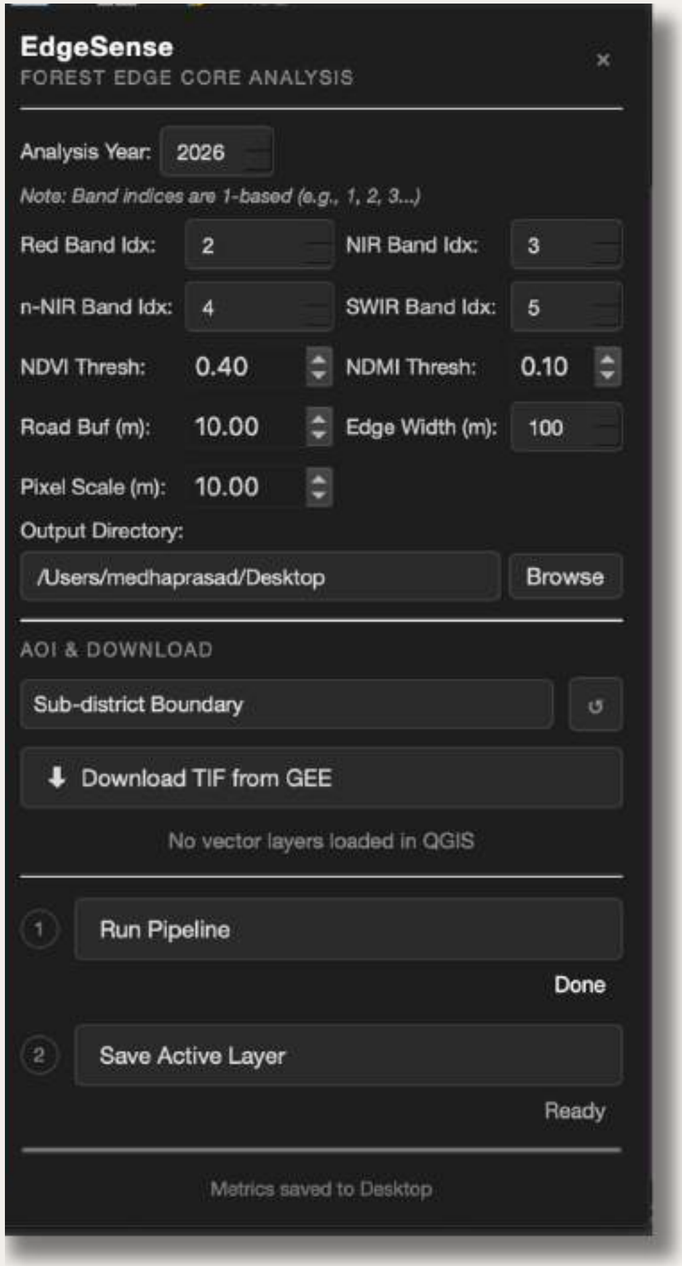
KEY FEATURES

- **In-Platform AOI Preparation:** Users can add any standard .shp file and select the Area of Interest inside QGIS.

- **Automated GEE Retrieval:** When the user clicks **"Download TIF from GEE",** the plugin **uses the selected AOI features**, builds a single geometry, sends it to Google Earth Engine, and **downloads the multi-band* GeoTIFF** for the **selected Analysis Year** (in a background thread so QGIS does not freeze). The **downloaded TIFF auto-loads into the workspace**.

- **Granular User Controls:** The Raster Toolbar **lets users adjust key parameters** such as **NDVI/NDMI thresholds, road buffer width (m),** and **pixel scale** before running the pipeline.

**We are using a standardized, hardcoded band configuration from our download scripts to ensure data consistency*



EDGESENSE

KEY FEATURES

- **Dynamic Raster Visualization:** During run_full_pipeline() (after user clicks **"Run Pipeline"**), **intermediate outputs** are written to temporary files and loaded as **QgsRasterLayer** objects. Users can visually inspect the NDVI, NDMI, Forest Mask, Road Mask, and Edge/Core Mask directly on the canvas.

- **Comprehensive Metric Extraction:** The plugin **automatically computes & exports spatial metrics** to a user selected output directory.

- **State-Aware UI & Exporting:** The interface **shows live progress. Users can** also **save** any **active intermediate raster** as a permanent GeoTIFF using **"Save Active Layer"**.

THANK YOU

Team: WFH

Github: EdgeSense